

## 03\_Class\_Activity

Bill Perry

### In class activity 3:



### What did we do last time?

- Implement data pipeline best practices
- Apply controlled vocabulary and naming conventions
- Create effective visualizations
- Customize plots for publication quality
- Combine multiple plots into composite figures

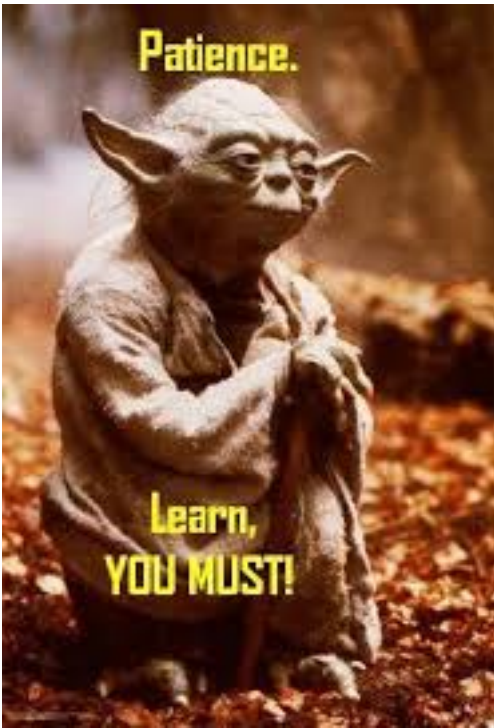
```
ggplot(name_df, aes(x_variable, y_variable, color = categorical_variable)) +  
#     dataframe, aesthetics(x and y variables, mapping of color or fill or shape) +  
  geom_point() +  
# this is the geometry you want and can add more layers like  
  geom_line()
```

- What questions do you have and what is unclear
- What did not work so far when you started the homework?

### Objectives and goals for today

#### Today's Objectives

1. Implement descriptive statistics in R
2. Calculate measures of central tendency and spread
3. Compare distributions of data from different groups
4. Create effective visualizations of descriptive statistics
5. Interpret the meaning of these statistics in a biological context



## Part 1: Setting Up Your Environment

First, let's load the necessary packages and import our data:

```
# Load required packages
```

```
library(knitr)           # For creating tables
library(moments)         # For calculating skewness and kurtosis
library(skimr)           # for summary stats
library(flextable)       # for tables if you want - now tinytable
library(tidyverse)       # For data wrangling and visualization
```

```
— Attaching core tidyverse packages ————— tidyverse 2.0.0 —
```

```
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2    3.5.2    ✓ tibble     3.3.0
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.1.0
```

```
— Conflicts ————— tidyverse_conflicts() —
```

```
* purrr::compose() masks flextable::compose()
* dplyr::filter()  masks stats::filter()
* dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Set a consistent theme for our plots
theme_set(theme_minimal(base_size = 12))
```

## Getting the data

## 💡 Practice Exercise 1: Loading and Examining the Grayling Data

We'll be working with data on arctic grayling fish from two different lakes (I3 and I8).

```
# Write your code here to read in the file
# How do you examine the data - what are the ways you think and lets try it!

# Load the grayling data
g_df <- read_csv("data/gray_I3_I8.csv")
```

```
Rows: 168 Columns: 5
— Column specification —————
Delimiter: ","
chr (2): lake, species
dbl (3): site, length_mm, mass_g

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# View the first few rows
head(g_df)
```

```
# A tibble: 6 × 5
  site lake species length_mm mass_g
<dbl> <chr> <chr>      <dbl> <dbl>
1  113 I3   arctic grayling    266    135
2  113 I3   arctic grayling    290    185
3  113 I3   arctic grayling    262    145
4  113 I3   arctic grayling    275    160
5  113 I3   arctic grayling    240    105
6  113 I3   arctic grayling    265    145
```

```
# Examine the data structure
glimpse(g_df)
```

```
Rows: 168
Columns: 5
$ site      <dbl> 113, 113, 113, 113, 113, 113, 113, 113, 113, 113, 113, 113, ...
$ lake      <chr> "I3", "I3", "I3", "I3", "I3", "I3", "I3", "I3", "I3", "I3", "I3", ...
$ species   <chr> "arctic grayling", "arctic grayling", "arctic grayling", "ar...
$ length_mm <dbl> 266, 290, 262, 275, 240, 265, 265, 253, 246, 203, 289, 239, ...
$ mass_g    <dbl> 135, 185, 145, 160, 105, 145, 150, 130, 130, 71, 179, 108, 1...
```

```
# Get a statistical summary
summary(g_df)
```

|          | site | lake             | species          | length_mm     |
|----------|------|------------------|------------------|---------------|
| Min.     | :113 | Length:168       | Length:168       | Min.:191.0    |
| 1st Qu.: | 113  | Class :character | Class :character | 1st Qu.:270.8 |

```
Median :118   Mode  :character   Mode  :character   Median :324.5
Mean   :116                                     Mean   :324.5
3rd Qu.:118                                     3rd Qu.:377.0
Max.   :118                                     Max.   :440.0
```

```
      mass_g
Min.   : 53.0
1st Qu.:151.2
Median :340.0
Mean   :351.2
3rd Qu.:519.5
Max.   :889.0
NA's   :2
```

```
# How many fish do we have from each lake?
```

```
g_df %>%
  count(lake)
```

```
# A tibble: 2 × 2
```

```
  lake      n
  <chr> <int>
1 I3      66
2 I8     102
```

## Questions to Consider:

1. What variables are in our dataset?
2. What are their data types?
3. Are there any missing values?
4. What is the range of fish lengths in our dataset?
5. How many fish were sampled from each lake?

## Part 2: Calculating Descriptive Statistics

Let's calculate various descriptive statistics for our data:

## 💡 Practice Exercise 2: Measures of Central Tendency

Let's recreate the basic histogram of fish lengths from our last class. Use the `sculpin_df` data frame that's already loaded.

```
# Write your code here to read in the file
# How do you examine the data - what are the ways you think and lets try it!
# Calculate the mean and median fish length
mean(g_df$length_mm)
```

```
[1] 324.494
```

```
median(g_df$length_mm)
```

```
[1] 324.5
```

```
# Calculate mean and median by lake
g_df %>%
  group_by(lake) %>%
  summarise(
    mean_length = mean(length_mm),
    median_length = median(length_mm)
  )
```

```
# A tibble: 2 × 3
  lake mean_length median_length
<chr>    <dbl>         <dbl>
1 I3      266.           266
2 I8      363.           373
```

## Summarizing data - two ways

lets say we want to summarize the data and need to get n, means, standard deviation, standard error

We could do the following - if we had missing cells the code below would give an error

```
mean(g_df$length_mm)
```

```
[1] 324.494
```

```
mean(g_df$length_mm, na.rm = TRUE) # removes missing values
```

```
[1] 324.494
```

```
length(g_df$length_mm)
```

```
[1] 168
```

- the length counts missing and non-missing data
- however this would get old if we had to do this for everything and then to do it for the different groupings - lee and windward...

## We need to learn to pipe

passes things from the dataframe to a command and so on...

- the dataframe -> pipe command that feed the dataframe into -> next command

```
g_df %>% summarize(mean_length = mean(length_mm, na.rm = TRUE))
```

```
# A tibble: 1 × 1
  mean_length
    <dbl>
1       324.
```

## What is cool is we can do a lot of different things now

```
g_df %>%
  summarize(
    mean_length = mean(length_mm, na.rm = TRUE),
    sd_length = sd(length_mm, na.rm = TRUE),
    n_length = n())
```

```
# A tibble: 1 × 3
  mean_length sd_length n_length
    <dbl>      <dbl>    <int>
1       324.       65.0      168
```

## Super cool code in case there are missing values

```
g_df %>%
  summarize(
    mean_length = mean(length_mm, na.rm = TRUE),
    sd_length = sd(length_mm, na.rm = TRUE),
    n_length = sum(!is.na(length_mm)))
```

```
# A tibble: 1 × 3
  mean_length sd_length n_length
    <dbl>      <dbl>    <int>
1       324.       65.0      168
```

## Now for Spread...

### 💡 Practice Exercise 3: Measures of Spread

```
# Write your code here to read in the file
# Calculate standard deviation and variance
mean_length <- mean(g_df$length_mm, na.rm=TRUE)
sd_length <- sd(g_df$length_mm)
var_length <- var(g_df$length_mm)
sd_length
```

```
[1] 65.00659
```

```
var_length
```

```
[1] 4225.856
```

### 💡 Exercise 4: Calculate Quartiles and Percentiles

```
# Calculate quartiles for overall data
quartiles <- quantile(g_df$length_mm, probs = c(0.25, 0.5, 0.75))
# cat("First quartile (Q1):", quartiles[1], "mm\n")
# cat("Second quartile (Median):", quartiles[2], "mm\n")
# cat("Third quartile (Q3):", quartiles[3], "mm\n")

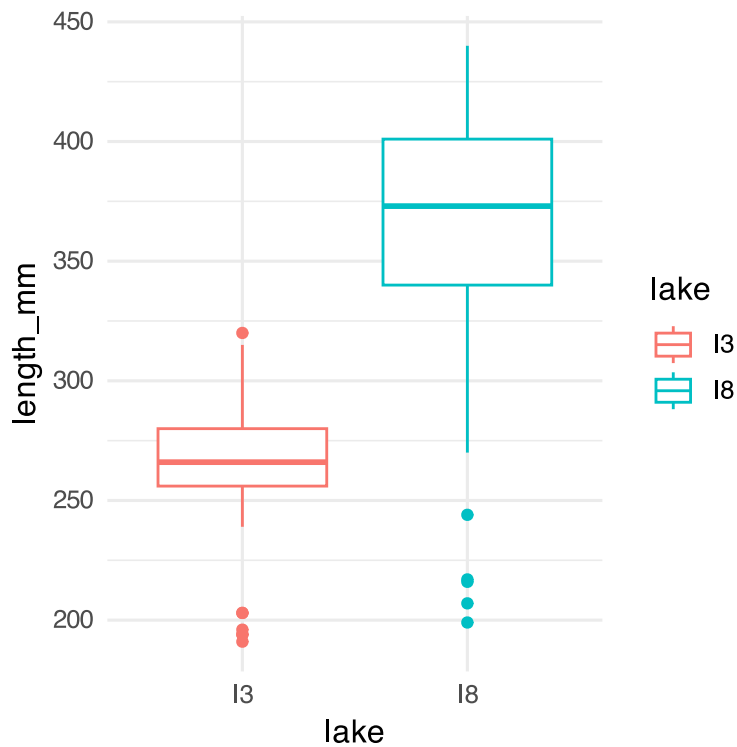
# Calculate a more comprehensive set of percentiles
percentiles <- quantile(g_df$length_mm,
                        probs = c(0.1, 0.25, 0.5, 0.75, 0.9))

# Display the percentiles using flextable
data.frame(
  Percentile = c("10th", "25th (Q1)", "50th (Median)", "75th (Q3)", "90th"),
  Value = percentiles
)
```

|     | Percentile    | Value  |
|-----|---------------|--------|
| 10% | 10th          | 251.10 |
| 25% | 25th (Q1)     | 270.75 |
| 50% | 50th (Median) | 324.50 |
| 75% | 75th (Q3)     | 377.00 |
| 90% | 90th          | 408.60 |

**Note you could add a box plot by lake to see this if you wanted**

```
g_df %>%
  ggplot(aes(lake, length_mm, color= lake))+
  geom_boxplot()
```



### 💡 Exercise 5: Calculate the Coefficient of Variation

The coefficient of variation (CV) is the standard deviation expressed as a percentage of the mean:

$$CV = \frac{s}{\bar{Y}} \times 100\%$$

```
# Calculate coefficient of variation
sd_length / mean_length * 100
```

```
[1] 20.03321
```

```
# Calculate by lake
g_df %>%
  group_by(lake) %>%
  summarise(
    mean_length = mean(length_mm),
    sd_length = sd(length_mm),
    cv_length = sd_length / mean_length * 100
  ) %>%
  flextable()
```

| lake | mean_length | sd_length | cv_length |
|------|-------------|-----------|-----------|
| l3   | 265.6061    | 28.30378  | 10.65630  |
| l8   | 362.5980    | 52.33901  | 14.43444  |

### Questions to Consider:

1. How do the means and medians compare within each lake? What might this tell you about the distribution?



2. Which lake has more variable fish lengths? How can you tell?
3. Why might the coefficient of variation be useful when comparing variability between different measurements (e.g., length vs. mass)?

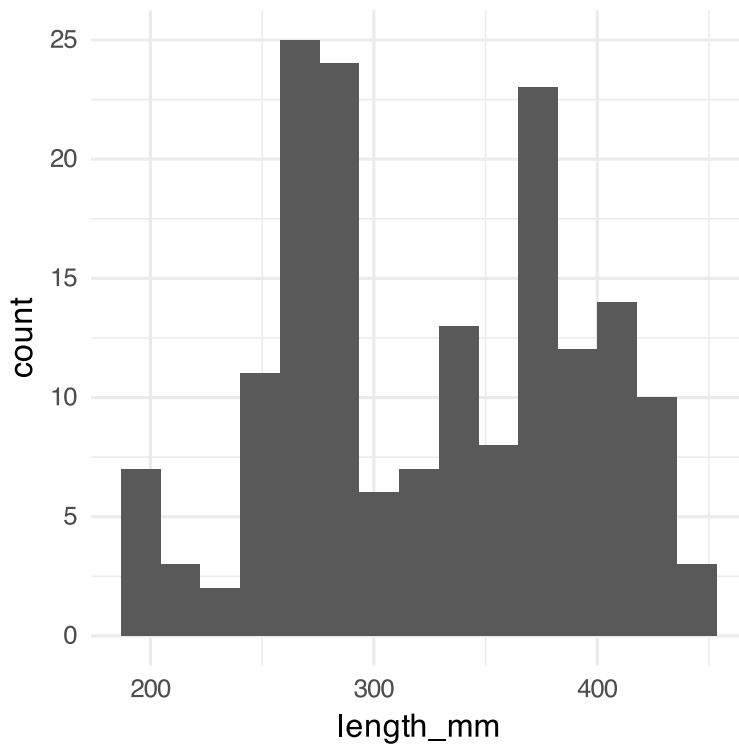
### Part 3: Visualizing Distributions

Visualizations can help us better understand the descriptive statistics we've calculated.

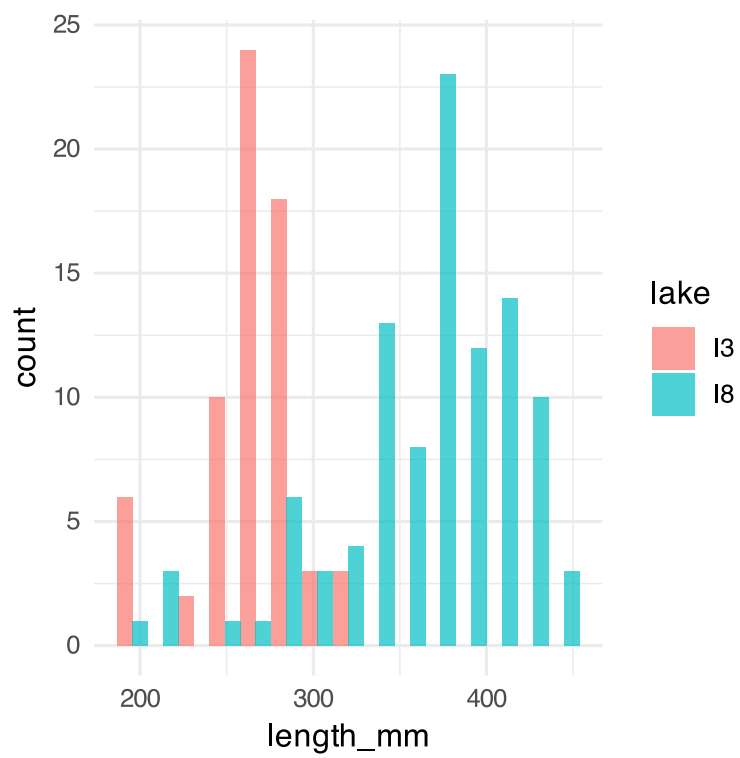
#### 💡 Exercise 6: Creating Histograms

One of the best ways to look at data is a histogram - and we will do it again

```
# Create a histogram of all fish lengths
g_df %>% ggplot(aes(x = length_mm)) +
  geom_histogram(bins = 15)
```



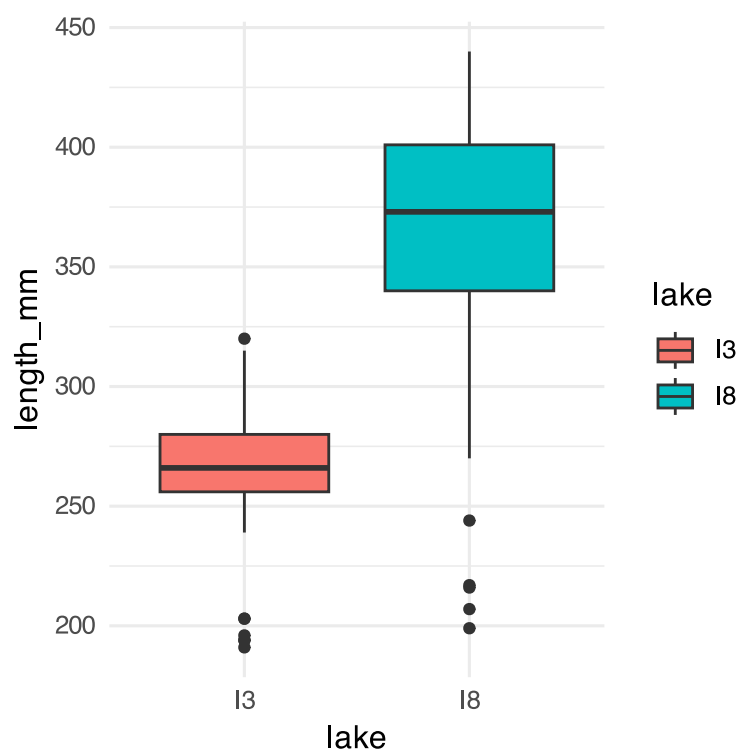
```
# Create histograms by lake
g_df %>% ggplot(aes(x = length_mm, fill = lake)) +
  geom_histogram(bins = 15, position = "dodge", alpha = 0.7)
```



## 💡 Exercise 7: Creating Box Plots

Personally I like box plots

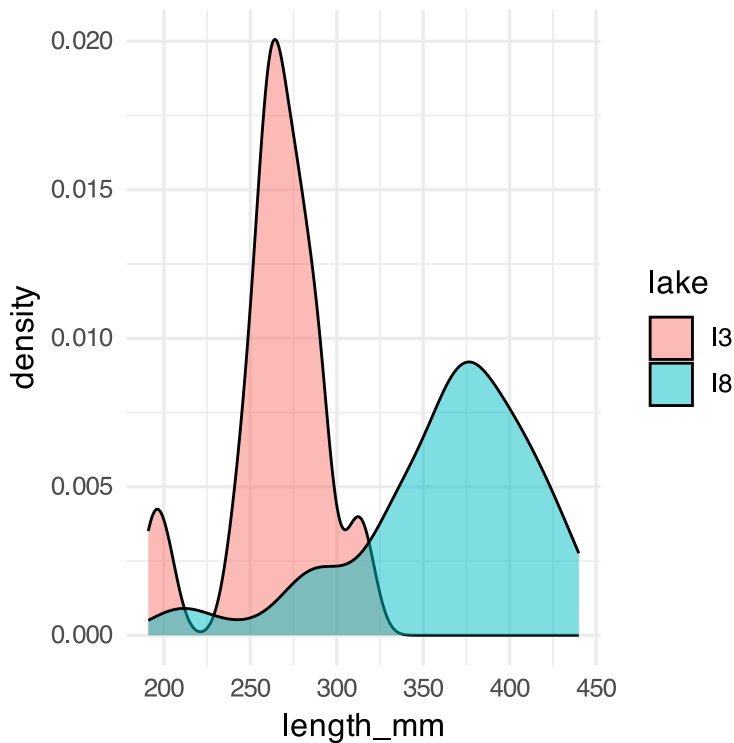
```
# Create a box plot comparing fish lengths by lake  
# Create a box plot comparing fish lengths by lake  
g_df %>% ggplot( aes(x = lake, y = length_mm, fill = lake)) +  
  geom_boxplot()
```



## 💡 Exercise 9: Creating Density Plots

Now these will be really important later on

```
## Create density plots
g_df %>% ggplot(aes(x = length_mm, fill = lake)) +
  geom_density(alpha = 0.5)
```



### Questions to Consider:

1. Which visualization best shows the differences in fish lengths between lakes?
2. What can you learn from the violin plots that might not be apparent from the box plots?
3. How would you interpret the cumulative frequency distribution?
4. What patterns or insights can you identify from these visualizations?

### Part 4: Interpreting the Results

Based on our analysis, we can make the following observations:

1. **Lake Differences:** Fish from Lake I8 are generally larger than those from Lake I3, both in length and mass.
2. **Variability:** Lake I8 shows greater variability in fish lengths and masses than Lake I3, as indicated by higher standard deviations and IQRs.
3. **Distribution Shape:**
  - Lake I3 fish lengths are more symmetrically distributed.
  - Lake I8 fish lengths show a slight negative skew, suggesting a few smaller fish pulling the distribution to the left.
4. **Length-Mass Relationship:** Both lakes show a strong positive correlation between fish length and mass, following an approximately cubic relationship (mass increases with the cube of length).

## Guided Questions for Deeper Understanding of descriptive statistics

1. **Biological Interpretation:** What ecological factors might explain the differences in fish size between the two lakes?
2. **Statistical Reasoning:** Why might we prefer to use the median and IQR instead of the mean and standard deviation in some cases?
3. **Data Visualization:** Which visualization method was most effective for comparing the two lakes? Why?
4. **Scientific Communication:** How would you concisely summarize these findings in a scientific paper?
5. **Further Analysis:** What additional analyses might be useful to better understand this dataset?